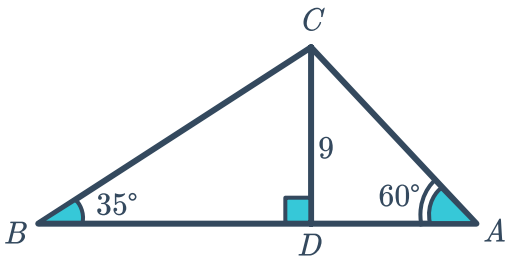
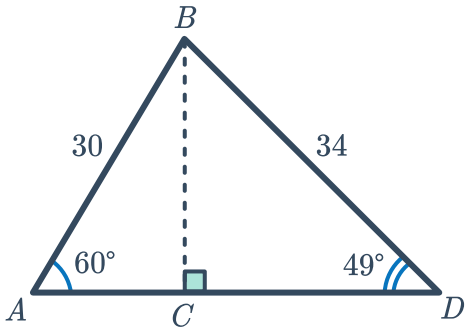
Unknown side lengths

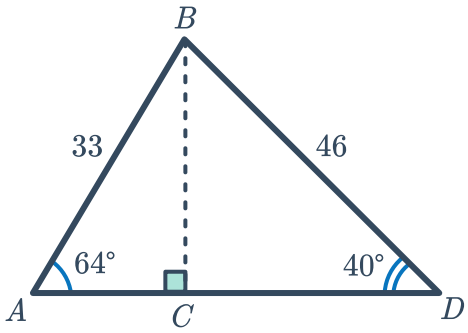
1 Consider the diagram shown:
Find the length of AB, correct to two decimal places.



2 Consider the following diagram:
Find the length of AD to two decimal places.

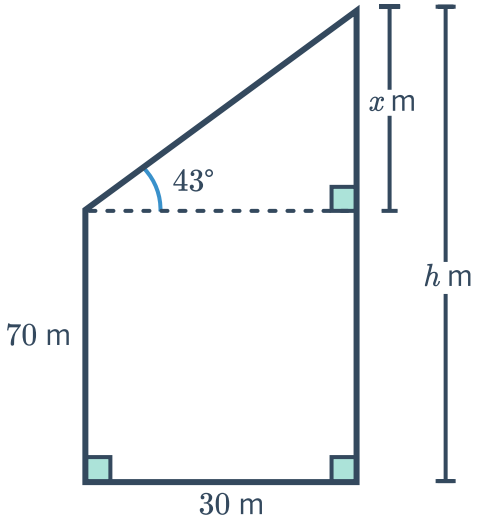


3 Consider the following diagram:
Find the length of AD to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Consider the following diagram:
Find the values of x and h to the nearest whole number.



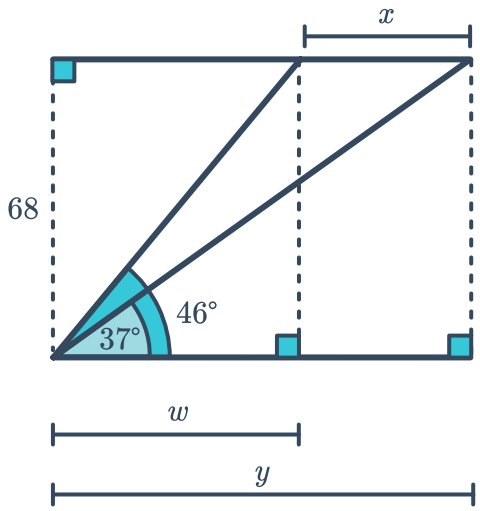
5 For each of the following diagrams.:

i Find y, correct to two decimal places.

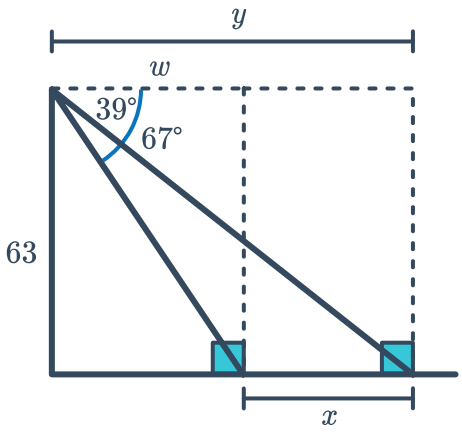
ii Find w, correct to two decimal places.

iii Hence, find the value of x, correct to one decimal place.

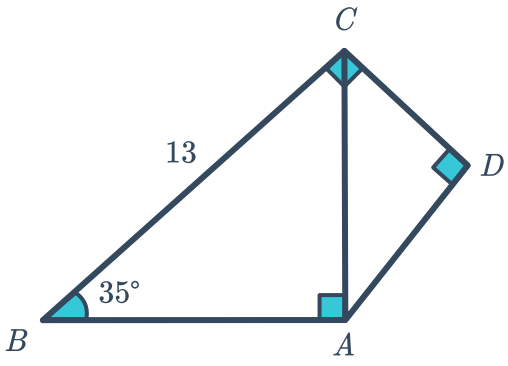
a



b

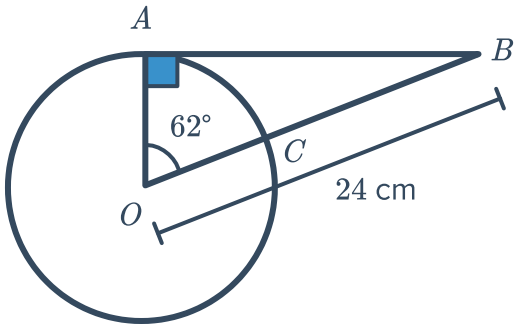


6 Consider the given diagram:
Find the length of CD, to one decimal place.

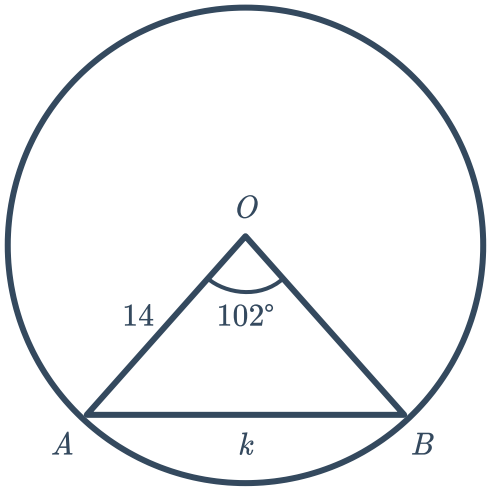


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 AB is a tangent to a circle with centre O. OB is 24 cm long and cuts the circle at C. Find the length of BC to the nearest centimetre.



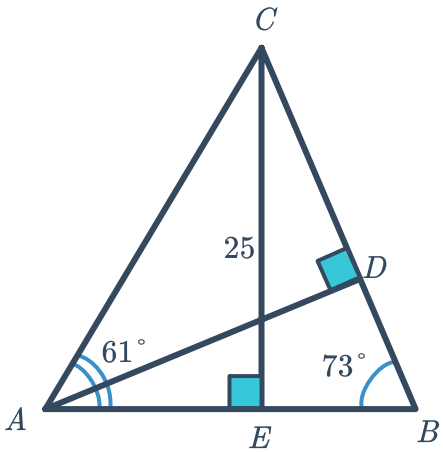
8 Consider the following diagram:
Find the value of k. Round your answer to two decimal places.



9 In the following diagram, $\angle CAE = 61^\circ$, $\angle CBE = 73^\circ$ and $CE = 25$.

a Hence, find the length of AB, correct to two decimal places.

b Find the length of BD, correct to one decimal place.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

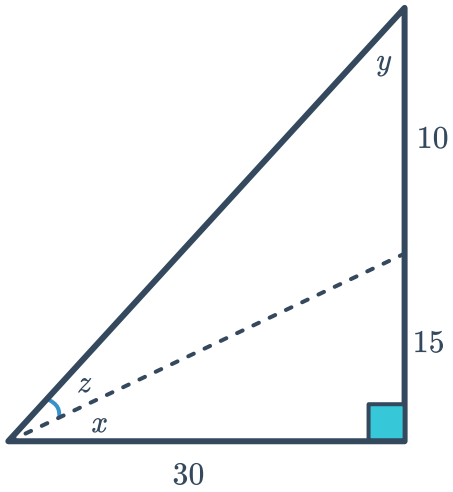
Unknown angles

10 Consider the given figure:

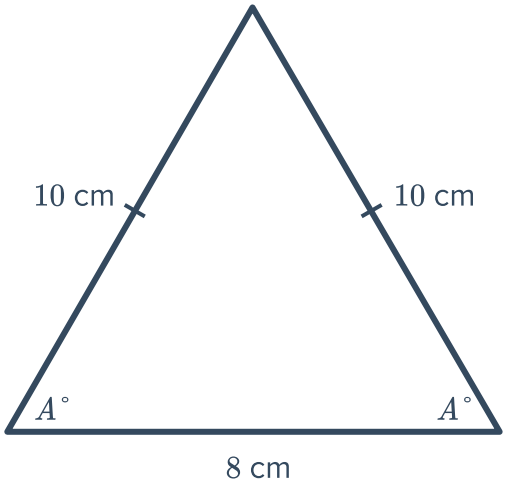
a Find x, correct to two decimal places.

b Find y, correct to two decimal places.

c Find z, correct to two decimal places.



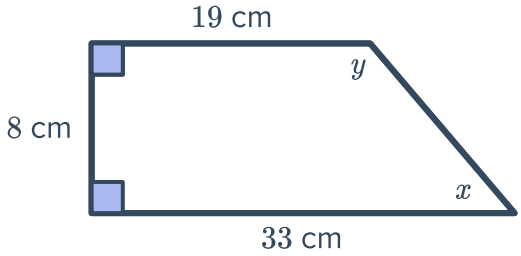
11 An isosceles triangle has equal side lengths of 10 cm and a base of 8 cm as shown. Calculate the size of angle A to one decimal place.



12 Consider the following figure:

a Find the size of angle x in degrees, correct to two decimal places.

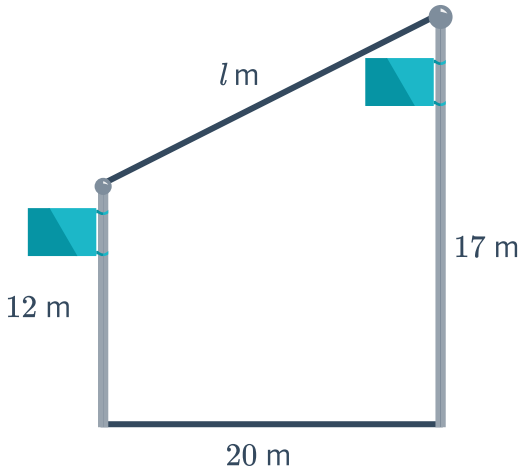
b Find the size of angle y in degrees, correct to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

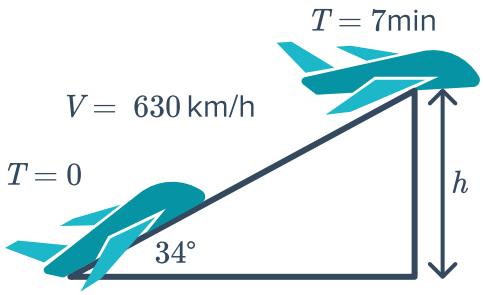
13 Two flag posts of height 12 m and 17 m are erected 20 m apart.
Find the length, l, of the string needed to join the tops of the two posts. Round your answer to one decimal place.



14 A jet takes off and leaves the runway at an angle of 34°. It continues to fly in this direction for 7 min at a speed of 630 km/h before levelling out.

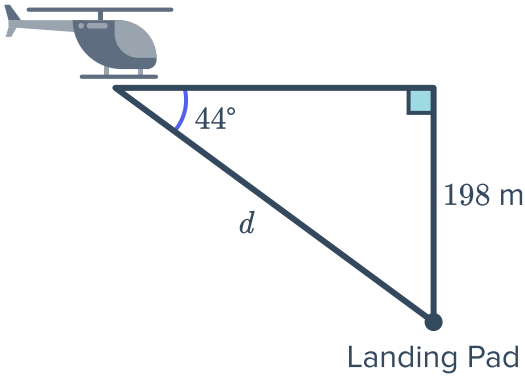
a Find the distance in metres covered by the jet just before levelling out.

b If the height of the jet just before levelling out is h m, calculate h.
Round your answer to the nearest metre.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

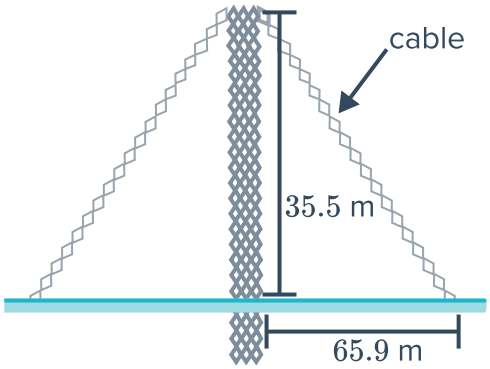
15 A helicopter is flying at an altitude of $198\ \text{m}$. Its landing pad is at an angle of depression of 44°.
Determine the distance, d, between the helicopter and the landing pad. Round your answer to the nearest whole number.



16 A suspension bridge is being built. The top of the concrete tower is $35.5\ \text{m}$ above the bridge and the connection point for the main cable is $65.9\ \text{m}$ from the tower. Assume that the concrete tower and the bridge are perpendicular to each other.

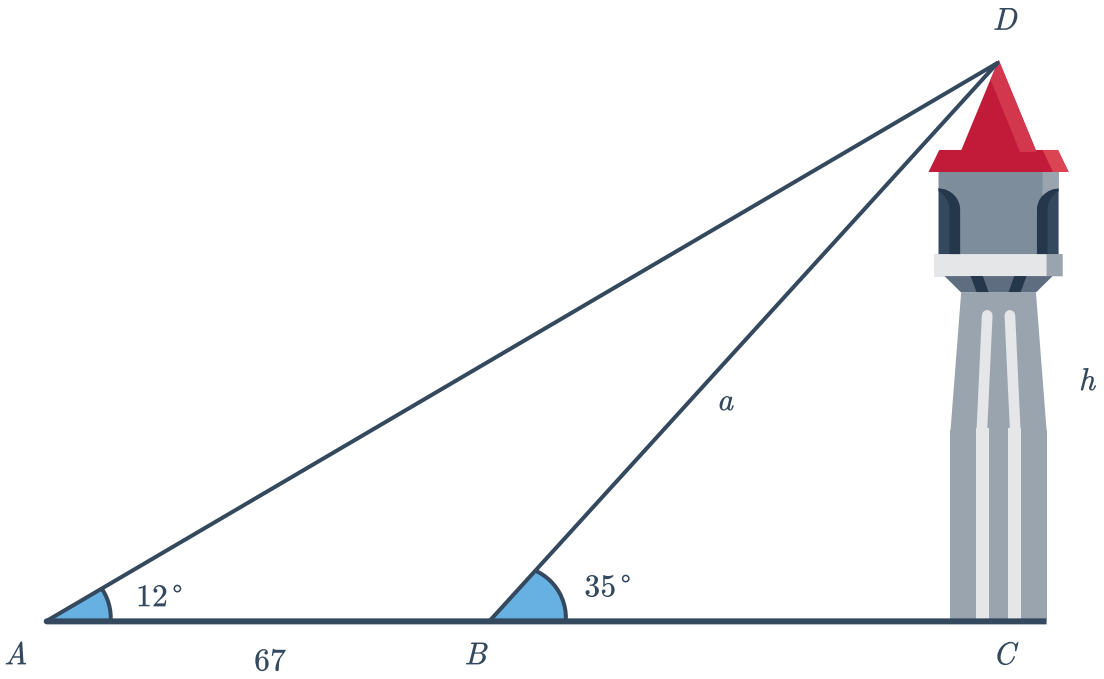
a Find the length of the cable to two decimal places.

b Find the angle the cable makes with the road to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

17 Mae observes a tower at an angle of elevation of 12°. The tower is perpendicular to the ground. Walking 67 m towards the tower, she finds that the angle of elevation increases to 35° as shown in the diagram below:

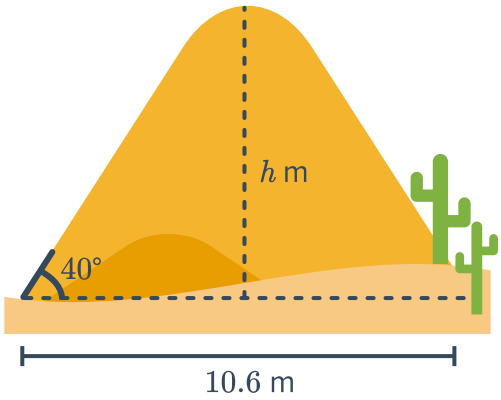


a Calculate the size of $\angle ADB$.

b Find the length of the side a. Round your answer to two decimal places.

c Find the height, h, of the tower. Round your answer to one decimal place.

18 A sand pile has an angle of 40° and is 10.6 m wide.
Find the height of the sand pile, h, to one decimal place.

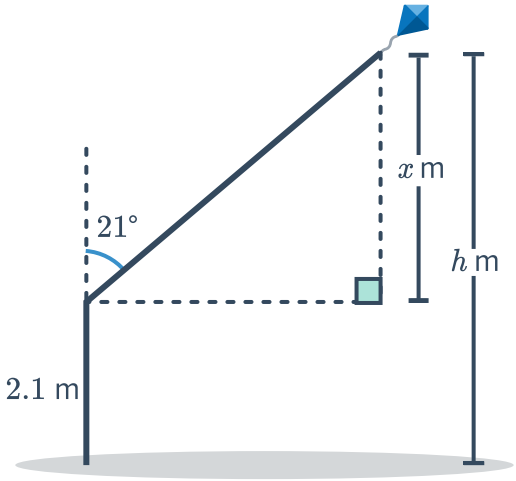


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 A girl is flying a kite that is attached to the end of a $23.4\ \text{m}$ length of string. The angle between the string and the vertical is 21°. If the girl is holding the string $2.1\ \text{m}$ above the ground:

a Find x, correct to two decimal places.

b Hence find the height h of the kite above the ground, correct to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647